

Pioneering Cooperative Air-Ground Instant Delivery using UAVs and Crowdsourced Couriers

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Instant delivery, which has a strict time limit, has become a fundamental service in modern cities. Increasing demands and labor shortages make efficient instant delivery more challenging. Recently, the industry has recruited Unmanned Aerial Vehicles (UAVs) and part-time couriers to address this critical issue. However, the inherent delivery ability of crowdsourced couriers and the limited battery capacity of UAVs make them hard to solely meet the rapidly increasing instant delivery demands. Therefore, this paper proposes the first air-ground cooperative UAV and Courier Delivery paradigm, called UCD, to develop an optimal courier recruitment plan and assignment of instant delivery tasks for UAVs and couriers. In the UCD paradigm, courier and UAV delivery models are introduced, and a recruitment plan that considers both delivery demand and courier availability is designed. With this foundation, a data-driven algorithm is proposed to optimize delivery task assignments, maximizing the flexibility of UAVs and the capacity of couriers. Comprehensive evaluations are conducted on multiple long-term real-world datasets to demonstrate the superior performance of UCD.

CCS Concepts: • **Human-centered computing** → **Scenario-based design**; **Ubiquitous and mobile computing design and evaluation methods**; **Human computer interaction (HCI)**.

Additional Key Words and Phrases: UAV, Courier, Data-driven, Air-Ground Instant Delivery, Cooperative Delivery

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1 INTRODUCTION

Instant delivery, which is a type of logistics service that requires delivering goods to the recipient as soon as possible, has seen remarkable growth in recent years. According to the analysis of Edision [67], the instant delivery market has increased by 4× to 7× from 2018 to 2021, with projections indicating continued growth. In China, the volume of takeout food deliveries is anticipated to surge at an annual rate of 28%, surpassing 100 billion by 2026 [37]. Globally, the takeout food market is predicted to double from 2022 to 2032, boasting an annual growth rate of 8.3% [36]. Unfortunately, the surge in instant delivery orders has outpaced the service capacity of couriers, leading to a decline in service quality. This is reflected in longer delivery time and a decrease in the overall order response rate. For example, online food delivery orders from a large number of restaurants across the USA have exceeded their service capacity, and the phenomenon has lasted for months [7]. In the winter of 2022 in Shanghai, one of the largest cities in China, a large number of instant parcels can not be delivered on time [59].

To guarantee the enhanced service quality, various novel delivery modes are being explored by academic and industrial communities. Among these, the most promising approaches involve crowdsourced delivery [24, 27, 29] and Unmanned Aerial Vehicle (UAV) delivery [17, 30, 42, 53]. Specifically, crowdsourcing delivery allows the express company to recruit part-time couriers to carry out deliveries [2, 15]. The flexible working hours attract individuals to participate in instant delivery during their free time, which can help enhance the overall efficiency of the delivery process. Some economic practices have also been achieved, such as Sweden's MyWays [50], America's TaskRabbit [48], and China's Dada Now [33] and Fengniao [27]; all of these leverage crowdsourcing delivery as their primary delivery method. Delivery UAVs, which can be remotely controlled by both intelligent algorithms and remote pilots, are promising to enable faster instant delivery at a much lower delivery cost [64]. UAV delivery can also bring an advantage in energy consumption and pollution emission reduction [82]. Specifically, the research of the life-cycle energy consumption estimation of UAV delivery [61] shows that the energy consumption of delivery UAVs is only 41% and 48% of that of the fossil truck and gas truck, respectively. Thus, UAV delivery can significantly reduce the greenhouse gases such as CO_2 , SO_2 and NO_x [28, 39]. Except for UAV-based delivery, industry has applied Electric Vehicles (EVs) based parcel-shipping methods to tackle to environmental concern caused by fossil vehicles. The comparison of environmentally friendly between UAVs-based and EVs-based delivery methods is to be investigated in future research. Motivated by such economic and environmental advantages of UAVs, the UAVs delivery mode is worth studying. Meituan company in China has started UAV instant delivery in Shanghai [10] and Shenzhen [57], and has safely delivered more than 180 thousand instant parcels such as food and coffee till the end of 2023.

However, the practical applications of crowdsourced delivery and UAV delivery reveal certain limitations that decline the service efficiency of instant delivery services. Firstly, the delivery capacity of crowdsourcing is dynamic and unstable [58, 66], since the service willingness of part-time individuals is easily changed due to time, date, weather and etc, [26, 32]. This dynamic and unstable delivery capacity can significantly degrade the service in terms of overall delivery capacity and average delivery time for each order. Secondly, UAVs face challenges in consistently delivering parcels to designated destinations, particularly when handling heavy items or covering long distances, primarily due to their limited battery capacity. The presence of no-fly zones further restrict the potential wider application of UAV delivery [72, 76].

Considering the limitations of independently leveraging crowdsourced delivery and UAV delivery, this paper explores an innovative company-oriented cooperative air-ground working paradigm, whose basic idea is illustrated

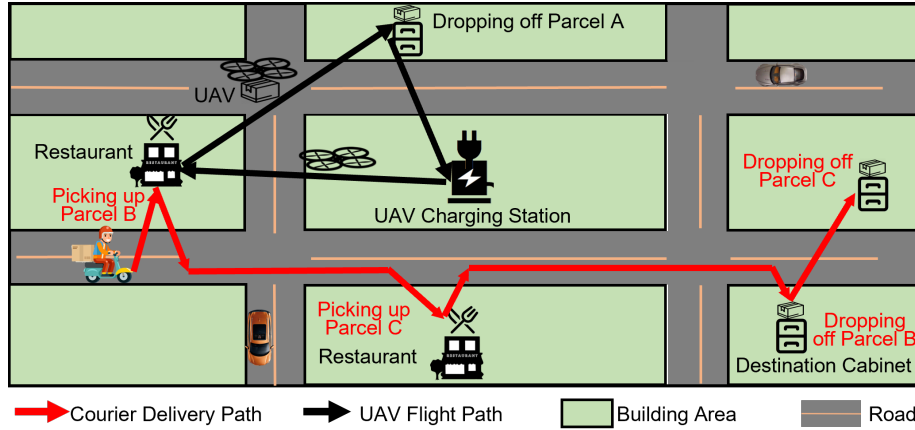


Fig. 1. An illustration of cooperative air-ground instant deliveries.

in Fig. 1 through an instance. Specifically, the express company assigns the UAV to deliver the nearby parcel A while assigning parcels B and C to the same courier, since the destinations of parcel B and C are near. This approach improves global delivery efficiency. Furthermore, to adapt to the time-varying instant delivery order distribution, we design a supervised learning-based neural network module to predict the order distribution in advance, followed by a courier number estimation model to determine the optimal number of couriers that should be employed to ensure the requirements of future orders. Subsequently, an online air-ground cooperative scheduling algorithm is built to swiftly assign orders to UAVs or couriers. Evaluation conducted on real-world datasets affirms the efficiency of our approach. Specifically, our approach strategically recruits a significantly reduced number of couriers, thus not only saving operational costs but also alleviating the negative impacts of the vulnerable delivery capacities of available couriers. Moreover, our approach enables the successful completion of a higher volume of instant delivery orders within the required time window by facilitating coordination between UAVs and couriers.

The contribution of this paper can be summarized as follows:

- To the best of the authors' knowledge, this is the first work on the company-oriented cooperative instant delivery paradigm using both UAVs and crowdsourced couriers. Through in-depth data-driven analysis on real-world data, some practical issues are observed to guide the design of the cooperative delivery framework UCD for both the appropriate number of recruited couriers and delivery tasks allocation between UAVs and couriers.
- In UCD, the courier delivery model and the UAV delivery model are first presented; then, a delivery demand- and courier-aware recruitment plan is designed. Based on this, a data-driven algorithm is designed to make appropriate delivery task assignment plan, which fully takes advantages of the flexibility of UAVs and the delivery capacity of couriers.
- Using a one-month delivery dataset and the corresponding contextual dataset, the performance of UCD is comprehensively evaluated. The evaluation results demonstrate that UCD significantly outperforms the baselines in terms of multiple metrics including the total delivered number, the required courier number, and the earned reward of crowdsourced couriers.

The remaining part of the paper is organized as follows. Sec.2 reviews the literature related to both crowdsourced delivery and UAV delivery. Sec.3 conducts a preliminary real-world data analysis to describe the motivation and

problems of this paper. Sec.4 presents the cooperative model involving crowdsourced couriers and UAVs. Sec.5 formulates the optimization problem and designs the corresponding solution. The performance of the cooperative delivery paradigm is evaluated in Sec.6, followed by the conclusion in Sec.7.

2 RELATED WORK

2.1 Crowdsourcing Delivery

One obvious advantage of crowdsourcing delivery is the large pool of potential couriers; namely, anyone with a vehicle (e.g., private car, taxi, or bike) can apply to become a courier. Devari et al. [14] demonstrated that using the private cars of nearby friends in social networks can provide cost-effective, speedy, and reliable delivery. Ding et al. [15] investigated parcel delivery by crowdsourced couriers using public transport. Dupljanin et al. [24] studied the performance of crowdsourcing delivery using various types of vehicles, including private cars, bicycles, and taxis. Liu et al. [44] distinguished the taxi-based crowdsourcing delivery into two situations according to the status of the taxis being occupied by passengers or not. And the authors respectively proposed algorithms for task allocation and path planning for delivery. Fatehi et al. [26] combined crowdsourcing with robust queuing and routing theories to satisfy instant delivery orders. Ta et al. [63] introduced the foundations of Self-Determination Theory and the heuristic-systematic model to examine how variations in messages presented to crowdsourced agents can enhance participation and performance outcomes.

However, part-time crowdsourced couriers generally exhibit a significant temporal-spatial preference for deliveries; namely, they prefer to deliver parcels within specific areas during particular times [1, 55]. Several studies have investigated ways to alleviate this limitation of crowdsourcing delivery. For example, Wang et al. [71] presented a crowdsourced recommending-and-grabbing instant delivery allocation system that considers the dynamics of crowdsourced couriers and allocates delivery orders accordingly. Sun et al. [62] examined the delivery route recommendation for crowdsourced couriers. However, the delivery dynamics of crowdsourced couriers significantly constrain their delivery capacity. For example, a study by Chen et al. [11] found that 14,000 taxis in a city can deliver only 10,000 to 20,000 parcels per day because many taxis frequently fail to find appropriate parcels to deliver.

2.2 UAV Delivery

To achieve efficient delivery, UAVs are expected to grasp and drop parcels automatically. The researchers designed and implemented grasping technologies for UAV delivery [9, 12]. These technologies allow UAVs to load and drop parcels without human intervention. The energy provisioning for UAVs is also an important issue for UAV delivery. Won et al. [73] designed a battery-swapping system for UAVs, which enables continuous and persistent delivery by UAVs. Wu et al. [74] studied to charge UAVs with the wireless charging technology. Currently, the maximum charging power reaches 500 watts [4], comparable to wired charging technology [19]. Dominguez et al. [20] proposed that widely deployed cellular infrastructure can contribute to charging the city-wide delivery UAVs as well as provide a robust wireless network to control the delivery UAVs in our cooperative UAV-courier delivery paradigm. If the cellular infrastructure method cannot be used for UAV charging, other charging station localization and deployment such as [35, 54, 69] can be utilized for UAV charging schedule. These works mainly optimize the locations of charging stations of the delivery UAVs so that the infrastructure cost is minimized and the coverage performance of the mission area is optimized. These works can be the foundation of the studied cooperative UAV-courier delivery paradigm. Well-established regulations are the foundation of the civil application of delivery UAVs[49]. For example, in Shenzhen, the first city to allow UAV for commercial instant delivery, established and published the UAV management regulations[51] in 2021, which dedicated set of regulatory for delivery UAVs [38]. The well-established regulations can help the stakeholders, such as the UAV manufacturers, express companies and the city managers, to prompt the applications of delivery UAVs.

Another line of UAV delivery research is the task and routing scheduling. Song et al. [60] studied the persistent delivery problem, in which each UAV 1) starts from a station and delivers a parcel, 2) returns to another station for charging and reloads a new parcel, and preparatory to serving additional customers. Delivery range is the most critical issue for UAV delivery. Gawel et al. [30] studied the UAV station location determination problem of delivery UAVs. When UAV swarms are used to cooperatively delivery, collision avoidance within UAVs is another challenge. One UAV can detect and recognize nearby obstacles using the Artificial Intelligence (AI)-based objective recognition from the image [5, 6], and then leverage a sampling-based algorithm to re-plan its flight route [43]. Considering the limited delivery range, a body of works has studied to route and relay UAVs with ground trucks [8, 21]. In case of avoiding using the fossil trucks, Yan et al. [52] and Choudhury et al. [13] proposed relaying UAVs using public buses to extend the delivery range of UAVs.

However, delivery UAVs face multiple limitations. First, the limited battery capacity limits their delivery range and delivery of some heavy parcels. Second, with more UAVs utilized for delivery, the low-altitude airspace would be congested and significantly limit the available UAVs [25, 41]. Third, due to public safety and privacy concerns, there are generally multiple no-fly zones [22, 23], where delivery UAVs are prohibited from flying.

2.3 Summary: Limitation of Existing Works and Technical Contribution of UCD

The practical applications of crowdsourced delivery and UAV delivery suffer some certain limitations, which may result in dramatic service degradation of the instant delivery service. Firstly, the easily changing service willingness of part-time individuals causes the dynamic and unstable delivery capacity of crowdsourced delivery [26, 32]. These existing solutions can alleviate such dynamic and unstable capacity but cannot completely avoid it. Besides, UAVs face challenges in maintaining consistent delivery, especially when handling heavy or long-distance parcels, primarily due to their constrained battery capacity. The safety and security concerns in the cities further restrict the potential application of UAV delivery [72, 76]. The advantages of UAVs and crowdsourcing enable them to make up for each other's limitations in instant delivery: the express company can maintain a UAV fleet to complete a considerable proportion of the instant deliveries in general case, thus saving a large amount of cost for maintaining a large group of human couriers. For the rest proportion of the undelivered parcels, due to no-fly zone restrictions, delivery order bursts during peak hours or heavy parcels, etc., part-time couriers can be recruited for the delivery.

From a technical perspective, the existing UAV route planning algorithms mainly consider the flight capacity, such as flight endurance and speed, and the parcel information such as weight, location and delivery deadline. While in UCD, inspired by the real-world food delivery dataset, the data-driven UAV-Courier cooperative delivery schedule is designed, including the appropriate recruited courier number determination and the delivery task assignment among UAVs and couriers. Especially, in the task assignment, the routes of couriers and UAVs would impact each other. Namely, when planning the routes of UAVs, the delivery model and the status of couriers should be considered; and when planning the routes of couriers, the delivery model and the status of UAVs should also be considered.

3 PRELIMINARY

This section first introduces the information of the data-sets, then conducts the feasibility, followed by the technical challenge analysis and problem statement of UCD. And the preliminary idea of the solution is introduced.

3.1 Dataset Description

Multiple datasets, the *aBeacon* dataset and two context datasets, are utilized for the design of UCD. Table. 1 summarizes the basic information of the datasets, which is introduced in detail.

Table 1. Basic Information of Datasets.

Datasets	# of couriers	Area(km ²)	# of Records	Sampling Interval
aBeacon	≥ 31K	6400	≥ 802K	1 – 3 min
Context-weather	N/A	6400	≥ 8K	30 min
Context-region	N/A	6400	≥ 180K	N/A

The aBeacon Dataset: Yang et al. [16] and Ding et al. [79] cooperated with the Alibaba Group, one of the biggest instant delivery companies in China, to study the instant delivery scheduling problem. The authors released a one-month dataset of the instant delivery in Shanghai. The original dataset consists of the delivery status data, the trace data and Bluetooth broadcast data of 31,131 couriers at 2,466 merchant locations during the month. More than 802,000 instant delivery orders within an area of 6,400 km² were successfully extracted from the trace data and the delivery status data. The daily order number reaches about 30,000, and the trace sampling interval is 1-3 minutes. The ordering time, delivery time, the restaurant location, and the receipt location are included in each order record. Fig. 2 demonstrates the heat-map of the pick-up locations of the parcels.

Two Contextual Datasets: To predict the delivery demand, the contextual datasets, i.e., the weather dataset and the functionality region dataset, from the same period as the aBeacon dataset in Shanghai, are collected. The weather dataset is crawled from the website [65], which updates the weather condition of the major cities of the world every 30 minutes. Each weather sample consists of the weather, temperature, humidity, wind velocity, and air pressure. The total weather context sample number used in this work is over 8,000. The functionality region data were collected from the OpenStreetMap [46]. The entire city is divided into 6400 square grids with the same length of 1km. Every grid consists of different kinds of buildings or infrastructures, which are labeled as one of the following categories: Leisure, Commercial, Tourism, Healthcare, Office, Residential, Study, or Industrial. The original functionality dataset includes over 180,000 records of raw data. Some works categorized a city grid into different functionalities using the labels within the grid [80], which utilizes the frequency of the labels to measure their importance and categorizes the functionality accordingly. Some other works combine the labels and the human mobility pattern data for grid functionality categorization [40, 68], in which the inter-region relations among grids contained in the human mobility data are extracted and utilized. Since it is non-trivial to obtain the human mobility data for the same city and period as the aBeacon data, due to the privacy and other concerns, this work leverages the label-based method for grid functionality categorization.

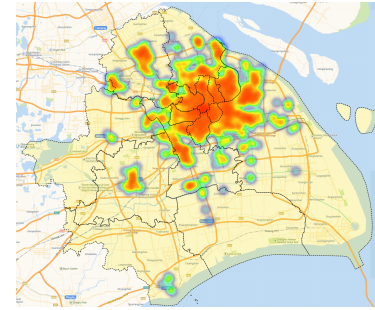


Fig. 2. Spatial Distribution of the Instant Delivery Pick-ups.

3.2 Data Analysis

Based on the real-world datasets, a data-driven analysis is conducted to reveal the temporal-spatial features of the instant delivery tasks, which motivates the solution design.

3.2.1 Limitation of Crowdsourcing Delivery. The spatial distribution of the pick-ups of the instant delivery orders from 12:00 to 12:30 is illustrated in Fig. 2, which shows that most instant deliveries occur in the northern area, the downtown of Shanghai. Besides, Fig. 3(c) shows the temporal distribution of the instant delivery orders with

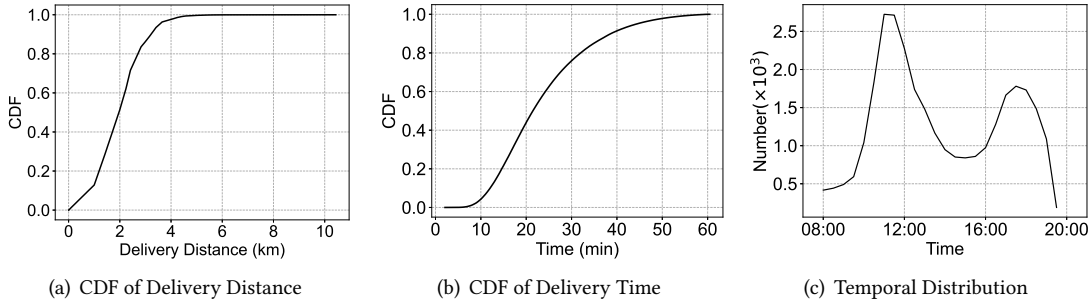


Fig. 3. Spatial and Temporal Information of the Instant Delivery.

every 30-minute interval in one day. There are two delivery peaks, i.e., the noon peak (10:30-13:00) and the afternoon peak (17:00-19:00). Most deliveries occur between 11:00 and 11:30, which is about 2,700 per day. Since one crowdsourced courier generally prefers to deliver within his/her familiar vicinity, there may be not enough couriers within the hot areas during the peak hours of the deliveries.

3.2.2 Feasibility and Limitation of UAV Delivery. Fig. 3(a) illustrates the delivery distance of an instant parcel, whose average delivery distance is 1.97km, and over 97.6% of instant deliveries are shorter than 4km. Since the flight distance of current commercial UAVs is as long as tens of kilometers, UAVs are capable of delivering most parcels. The energy consumption model of the UAV will be presented in Sec. 4.2 in detail. However, the total number of available UAVs is limited due to air traffic management, as stated in Sec. 1. There may not be enough UAVs to deliver all the parcels during peak hours and the UAVs cannot delay the deliveries to the off-peak hours, because the instant delivery is time-critical. Fig. 3(b) depicts the delivery time of dedicated instant couriers. The average delivery time from ordering to receipt of the instant delivery is 23.63 minutes, 24.1% of which are delivered over 30 minutes. Inspired by the high flexibility and speed, the UAVs can be much faster than current dedicated couriers.

3.3 Problem Statements

Inspired by the data-driven analysis above, this paper aims to address the delivery capacity gap of a limited fleet of UAVs by using crowdsourced part-time couriers. However, the part-time couriers on a crowdsourcing platform are open to anyone and may be recruited for other tasks. Thus, the express company needs to determine the appropriate number of couriers and request to recruit these couriers in advance on the crowdsourcing platform [84]. Thereafter, this paper aims to address the following two basic express company-oriented problems:

- **The courier number determination problem:** When recruiting the couriers in advance, the recruiting courier number should be appropriate. Requesting too few couriers may not fulfill the delivery demand and thus degrade the delivery performance; While requesting too many couriers would decrease the delivery task number of each courier, which not only brings extra recruitment cost but also harms the couriers' willingness to participate in the future tasks of the company.
- **The instant delivery task assignment problem:** The instant delivery task assignment between the UAVs and crowdsourced couriers needs to consider multiple factors including the delivery demand (i.e., delivery number, pick-up locations, and drop locations), the UAV status (i.e. location, available UAV number) as well as the couriers' delivery capacity determined by their locations and occupation status.

3.4 Solution Overview

With the problems stated above in mind, this paper proposes an air-ground cooperative instant delivery framework, namely UCD. The primary components of UCD are presented in Fig. 4, including the Data Module, the Cooperative Delivery Model Module, and the Cooperative Delivery Optimization Module.

The contextual data and the extracted delivery order data will first be input into UCD. Then, the delivery models of couriers and UAVs are presented in Sec. 4.1 and Sec. 4.2, respectively. With the contextual data and the delivery order data, a prediction model is designed to forecast the future instant delivery demand in Sec. 5.1. By combining the delivery models and the prediction results, a courier recruitment model is designed to employ an appropriate number of couriers in Sec. 5.2. Finally, the optimization problem of the delivery task assignment between UAVs and couriers is formulated in Sec. 5.3 and solved in Sec. 5.4.

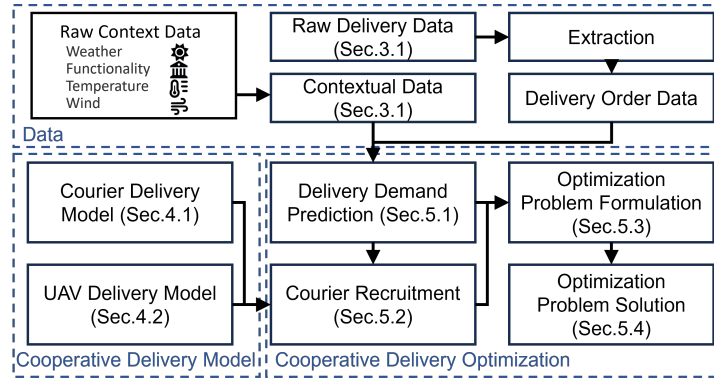


Fig. 4. The UCD Framework.

4 COOPERATIVE DELIVERY MODEL

4.1 Courier Delivery Model

Let $\mathbb{P}$ and $\mathbb{C}$ respectively indicate the sets of parcels ordered and the couriers delivering parcels. Let $\mathbb{P}(c) \subseteq \mathbb{P}$ represent the assigned parcel set to the courier $c \in \mathbb{C}$. The delivery path of $\mathbb{P}(c)$ is denoted as $T(c, \mathbb{P}(c)) = \{wp_i | i = [0, I]\}$, which consists of I waypoints wp with the starting waypoints being c 's current location wp_0 . i is the waypoint sequence along the path. wp_i can be a restaurant to pick up some parcels or a destination cabinet to drop off some parcels. wp_i is a 4-tuple: $(l_i, t_i, \mathbb{P}_i(c), \mu_i)$, in which l_i denotes the location, t_i is the time c arrives at wp_i , $\mathbb{P}_i(c)$ indicates the parcel set picked up or dropped off at wp_i . The binary parameter $\mu_i = 1$ indicates the parcel set $\mathbb{P}_i(c)$ being picked up at a restaurant; $\mu_i = -1$ denotes the parcel set $\mathbb{P}_i(c)$ being dropped off at a cabinet. Note that $\mathbb{P}(c)$ is the sum of the parcel set along $T(c, \mathbb{P}(c))$:

$$\mathbb{P}(c) = \{\cup \mathbb{P}_i(c) | \mu_i = 1\} = \{\cup \mathbb{P}_i(c) | \mu_i = -1\}. \quad (1)$$

Given a delivery path $T(c, \mathbb{P}(c))$, some constraints should be satisfied considering the delivery requirements of instant parcels:

- **The Courier Capacity Limit:** During delivery, the courier's delivery capacity, which is measured by both his simultaneously carried total parcel number and total parcel weight, should be limited. Let $N_{max}(c)$ and $W_{max}(c)$ respectively denote the maximum parcel number and maximum parcel weight of a courier. Since one courier only picks up or drops off parcels at a wp_i , let $\omega(p)$, $N_i(c)$ and $W_i(c)$ respectively indicate

the weight of parcel p , carrying parcel number and the weight of the courier c 's parcels at wp_i . Thus, the courier's carried parcel number limit is

$$N_i(c) \leq N_{max}(c), \forall i \in [0, I]. \quad (2)$$

$N_i(c)$ is updated when the courier reaches wp_i by

$$N_i(c) = \begin{cases} N_{i-1}(c) + |\mathbb{P}_i(c)|, & \text{if } \mu_i = 1, \text{ i.e., } wp_i \text{ is a restaurant} \\ N_{i-1}(c) - |\mathbb{P}_i(c)|, & \text{if } \mu_i = -1, \text{ i.e., } wp_i \text{ is a cabinet} \end{cases}, \quad (3)$$

where $|\mathbb{P}_i(c)|$ indicates the size of the set $\mathbb{P}_i(c)$. Meanwhile, the courier's carried parcel weight limit is

$$W_i(c) \leq W_{max}(c), \forall i \in [0, I], \quad (4)$$

where $W_i(c)$ is updated when the courier reaches wp_i by

$$W_i(c) = \begin{cases} W_{i-1}(c) + \sum_{p \in \mathbb{P}_i(c)} \omega(p), & \text{if } \mu_i = 1, \text{ i.e., } wp_i \text{ is at a restaurant} \\ W_{i-1}(c) - \sum_{p \in \mathbb{P}_i(c)} \omega(p), & \text{if } \mu_i = -1, \text{ i.e., } wp_i \text{ is at a cabinet} \end{cases}. \quad (5)$$

- **The Delivery Time Limit.** Since the instant delivery is time-critical, any parcel $p \in \mathbb{P}(c)$ should be delivered to its destination cabinet within Δt after being ordered. In this paper, Δt is assumed to be the same for all parcels, such as 40 minutes. When c is dropping off the instant parcel p at wp_i , the drop-off time of p is t_i and the order time of p is $o(p)$. Therefore, the delivery time limit is:

$$t_i \leq o(p) + \Delta t, \forall p \in \mathbb{P}_i(c) \subset \mathbb{P}(c). \quad (6)$$

For ease of representation, a delivery path of a courier $T(c, \mathbb{P}(c))$ is called a Feasible Courier Delivery Path if the definitions above are satisfied. Namely:

Definition 4.1 (Feasible Courier Delivery Path). A delivery path of couriers $T(c, \mathbb{P}(c))$ is feasible only if the courier can carry all the parcels at any waypoint, and every instant parcel can be shipped within the time limitation, namely Eq.(2), (4) and (6) are true.

If a parcel $p \in \mathbb{P}$ is assigned to the courier $c \in \mathbb{C}$, the new delivery path of c is $T(c, \mathbb{P}(c) \cup \{p\})$. If the delivery path $T(c, \mathbb{P}(c) \cup \{p\})$ is also feasible, the detour distance of c to deliver p can be computed as:

$$\Delta t(c, p, \mathbb{P}(c)) = \frac{\|T(c, \mathbb{P}(c) \cup \{p\})\|_c - \|T(c, \mathbb{P}(c))\|_c}{v(c)}, \quad (7)$$

where $\|T(c, \mathbb{P}(c))\|_c$ indicates the length of the courier trajectory and $v(c)$ is the velocity of the courier. Specifically, $v(c)$ may be different for different c because different courier may use different transportation tools.

If some parcels in the set $\mathbb{P}^+ \subseteq \mathbb{P}$ cannot be delivered by any courier in $\mathbb{C}$, extra couriers are recruited from the crowdsourcing platform to deliver these parcels. The set of newly recruited couriers is denoted as $\mathbb{C}^+$. Thus, the courier set will be updated as $\mathbb{C} = \mathbb{C} \cup \mathbb{C}^+$. Besides, if the couriers in the set $\mathbb{C}^- \subseteq \mathbb{C}$ have no parcel to deliver, they will be released to the crowdsourcing platform for other crowdsourcing tasks. The courier set thus is updated as $\mathbb{C} = \mathbb{C} \setminus \mathbb{C}^-$.

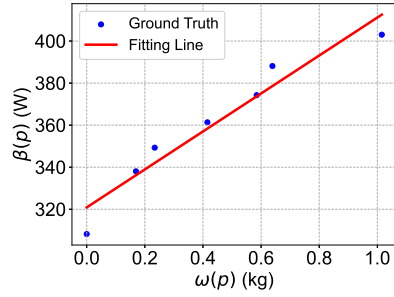


Fig. 5. UAV Energy Consumption Rate [18].

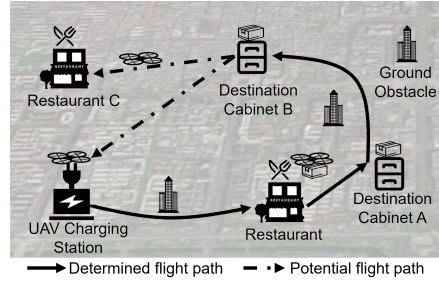


Fig. 6. Typical UAV Delivery Process[57, 75].

4.2 UAV Delivery Model

As for the UAV delivery, the energy consumption is the most critical, which is mainly determined by the parcel weight (determining the energy consumption rate) and delivery distance (determining the UAV flying time). This work takes DJI M210, a commercial UAV, as an example of the delivery UAV. Fig. 5 illustrates the energy consumption rate β (in Watts) of M210 with different parcel weights, in which the blue dots are the original energy consumption data measured by DJI [18]. The red curve is the linear fitting results, whose fitting function is:

$$\beta(\omega) = 90.3 \times \omega + 320.9, \quad (8)$$

where ω indicates the weight of the carried parcels of the UAV, 90.3 and 320.9 are the fitted weight and offset parameters by the linear fitting, respectively. Specifically, let $\beta(0)$ denote the energy consumption rate when the UAV carries no parcels. The average approximation fitting error to the original energy consumption data is only 1.7%. With the energy consumption model, we can obtain that M210 UAVs can fly for 34 minutes without a parcel, and for 25 minutes with a parcel of 1kg.

Let $\mathbb{P}(u)$ and $T(u, \mathbb{P}(u)) = \{wp_i | i \in [0, I]\}$ denote the delivery sequence of assigned parcel set to the UAV u and the flight path of u , respectively. $T(u, \mathbb{P}(u)) = \{wp_i | i \in [0, I]\}$ consists of I waypoints, each waypoint is 5-tuple: $(l_i, t_i, \mathbb{P}_i(u), \mu_i, E_i)$. The parameters l_i, t_i, μ_i , and $\mathbb{P}_i(u)$ are the same as those waypoints in $T(c, \mathbb{P}(c))$. E_i represent the remaining energy of u at wp_i . The typical UAV delivery process is shown in Fig. 6 [57, 75]:

- The assigned UAV $u \in \mathbb{U}$ first departs from its charging station and flies to the restaurant to pick up the instant parcels; then
- The UAV u generates the delivery path for carrying parcels $\mathbb{P}(u)$; after that,
- u flies to the destinations of the parcels according to the delivery sequence; finally,
- if there is no parcel deliverable by u , u will fly back to the charging station for energy replenishment; otherwise, u will be scheduled to deliver other parcels if its remaining energy $E(u)$ is sufficient.

In this case, u is allowed to pick up other parcels after delivering all its carried parcels if its remaining energy is sufficient, which prompts the delivery efficiency and turnover of UAVs. The delivery time of each parcel in $\mathbb{P}(u)$ should be shorter than the delivery time limit, similar to the Eq.(6): $t_i \leq o(p) + \Delta t, \forall p \in \mathbb{P}_i(u) \subset \mathbb{P}(u)$, where t_i and o_p respectively indicate the drop-off time at the wp_i and the order time of the parcel $p \in \mathbb{P}(u)$. The energy consumption of u of the whole delivery/flight path is:

$$\Delta E(u, \mathbb{P}(u)) = \beta(0) \frac{\|l_0, l_1\|_u}{v(u)} + \sum_{i=2}^{|\mathbb{P}(u)|} \beta(W_{i-1}(u)) \frac{\|l_{i-1}, l_i\|_u}{v(u)} + \beta(0) \frac{\|l_{I-1}, l_I\|_u}{v(u)}, \quad (9)$$

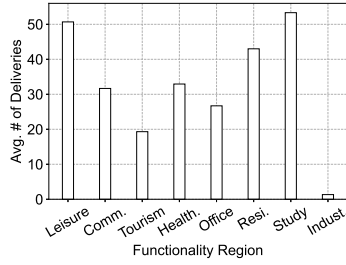


Fig. 7. Num. of Parcels in Each Grid.

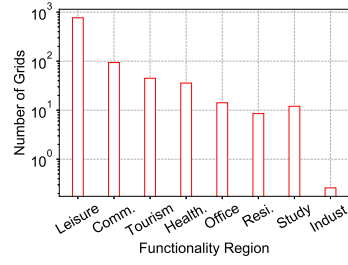


Fig. 8. Num. of Functionality Grids.

where $\|l_{i-1}, l_i\|_u$ is the flight distance from the $w_{p_{i-1}}$ to w_{p_i} ; v_u is the flight velocity of the UAV; $W_i(u)$ is the weight of the carried parcels of the UAV during different wps , which is updated similar to $W_i(c)$ in Eq.(5). Specifically, impacted by ground terrains (such as high buildings), the flight path $\|l_{i-1}, l_i\|$ from two wps may not be a straight line, which can be scheduled using the sampling-based algorithm [43], as appended in Appendix A.

Let $E(u)$ indicate the remaining energy of u before delivery. The condition in which u can deliver the instant parcels in $\mathbb{P}(u)$ is:

$$\Delta E(u, \mathbb{P}(u)) \leq \alpha E(u), \quad (10)$$

where α is smaller than 1, such as 0.8, and $(1 - \alpha)E(u)$ is the extra energy considering the uncertainty during flight. For example, the delivery UAV can detect the nearby birds or other UAVs with its on-board camera[78], and the it may need to iteratively re-plan its flight path to avoid collision, during which the sampling-based algorithm can also be used [43].

With the above descriptions, the feasible UAV delivery path $T(u, \mathbb{P}(u))$ is defined as follows, similar to the definition of the feasible courier delivery path.

Definition 4.2 (Feasible UAV Delivery Path). If a UAV u has sufficient energy to carry the parcels at any waypoint along the delivery path $T(u, \mathbb{P}(u))$, all parcels can be shipped within the delivery time limit, and then return to charging stations, the delivery path $T(u, \mathbb{P}(u))$, is a feasible delivery path of u .

Let p denote the assigned parcel to the UAV u . The delivery path of $T(u, \mathbb{P}(u) \cup \{p\})$ is first scheduled. The additional flying time of u for delivering p along the feasible delivery path $T(u, \mathbb{P}(u) \cup \{p\})$ is

$$\Delta t(u, p, \mathbb{P}(u)) = \frac{\|T(u, \mathbb{P}(u) \cup \{p\})\|_u - \|T(u, \mathbb{P}(u))\|_u}{v(u)}, \quad (11)$$

where $\|T(u, \mathbb{P}(u))\|_u$ is the flight distance of the delivery path.

5 COOPERATIVE DELIVERY OPTIMIZATION

To improve the delivery efficiency of the UAVs, a Multiple-Layer Perceptron (MLP) is leveraged to learn delivery occurrence patterns and predict the future delivery demand in Sec. 5.1. The predicted delivery demand is combined with the delivery capacities of UAVs and couriers to guide the courier recruitment in Sec. 5.2. The optimization problem is formulated in Sec. 5.3 and solved in Sec. 5.4.

5.1 Delivery Demand Prediction

Some previous works such as Guo et al.[34] and Zhang et al. [83] revealed that the weather conditions have a great impact on the instant parcel occurrence. Besides, the available datasets in Sec. 3.1 reveal that the area

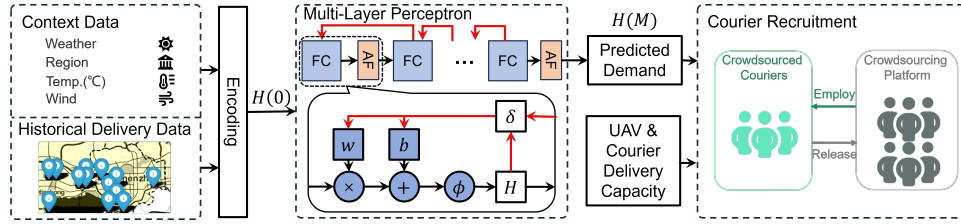


Fig. 9. Delivery Prediction and Courier Recruitment Architecture.

functionality plays an important role in the instant parcel occurrence as well. Specifically, the numbers of the parcels occurring at different functional grids is shown in Fig. 7. The average daily parcel numbers for the Study and Leisure grids are greater than 50, while the number for the Industrial grids is only 1.3 parcels. The daily parcel numbers of other functional grids vary from 19 to 43. Except for the number of parcels occurring in each kind of grid, the grid numbers of the 8 functionalities are shown in Fig. 8.

Since the average parcel number of the Industrial grids is negligible, the prediction model forecasts the potential delivery demand within other 7 kinds of functional grids.

The weather data and the area functionality data are input into the MLP as the contextual data for delivery demand prediction. Fig. 9 shows the architecture of the delivery demand prediction and the courier recruitment. The historical delivery data and contextual data from the first 23 days are used for model training, while the data for the following 7 days are used for validation. Let M denote the layer number in the MLP, and $H(j)$ indicate the output of j -th layer. The input of the MLP is $H(0)$ and $H(M)$ denotes the predicted parcel number, which is the output of the MLP. Specifically, the input $H(0)$ is

$$H(0) = [\psi, time, location, weather, functionality]^K, \quad (12)$$

where ψ_k denotes the occurrence instant parcel number in the k -th sample, the time and location respectively indicate the occurrence time of the ψ_k parcels. The details of the weather conditions and functionality regions refer to Sec. 3.1. $H(0)$ transmits along the black lines in the MLP as shown in Fig. 9. All layers in the MLP are Full-Connected layers (FC layers) with Activation Functions (AF). The output $H(j)$ of the j -th layer holds:

$$H(j) = \phi(H(j-1)W(j) + b(j)), \quad (13)$$

where $\phi(\cdot)$ is the activation function, $W(j)$ and $b(j)$ are respectively the weight parameter and the offset parameter of the j -th layer. To tackle the potential dead neuron [45], the Leaky-ReLU is leveraged as the AF in the MLP. Thus, $H(j)$ is

$$H(j) = \begin{cases} H(j-1)W(j) + b(j), & \text{if } H(j-1)W(j) + b(j) \geq 0 \\ 10^{-2} \times H(j-1)W(j) + b(j), & \text{otherwise} \end{cases} \quad (14)$$

To evaluate the prediction loss of the MLP, the Huber Loss is used as the loss function. Because it is not only differentiable everywhere, but also tolerates the outliers [47]. Besides, the Huber Loss strikes a balance between accuracy and robustness [31].

$$LOSS(y(j), \hat{y}(j)) = \begin{cases} \frac{1}{2}(y(j) - \hat{y}(j))^2, & |y(j) - \hat{y}(j)| \leq \delta \\ \delta|y(j) - \hat{y}(j)| - \frac{1}{2}\delta^2, & |y(j) - \hat{y}(j)| > \delta \end{cases}, \quad (15)$$

where $\delta = 1.0$ is the default value in PyTorch[56], $y(j)$ is the output of the j -th layer, and $\hat{y}(j)$ is the ground truth value in the j -th layer. We use the 5-fold cross validation to test the MLP. Specifically, $y(M)$ is the predicted

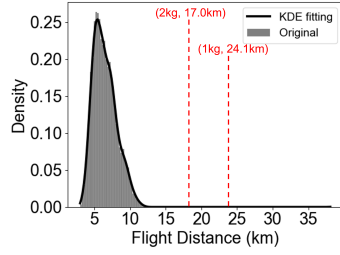


Fig. 10. PDF and KDE Fitting of Flight Distance of UAVs.

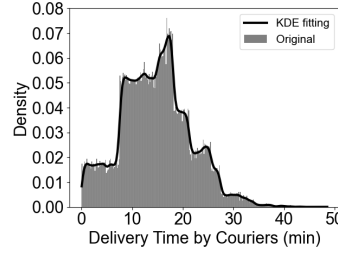


Fig. 11. PDF and KDE Fitting of Delivery Time by Couriers.

parcels of the MLP based on the contextual data in the validation set and $\hat{y}(M)$ is the historical parcel data in the validation set. The derived loss is back-propagated along the red lines in Fig. 9 and modifies the parameters of each layer. When the MLP is well-trained, we predict the possible deliveries in the 7-day contextual data, the predicted parcels are used for the courier recruitment in Sec. 5.2.

5.2 Courier Recruitment

The part-time couriers need to be recruited in advance to deliver the future parcels, because the available candidate couriers of a crowdsourcing platform are not only spatial-dynamic but limited [84]. To determine the appropriate recruited courier number, UCD needs to not only predict the delivery demand above, but also respectively estimate the delivery capacities of the UAVs and the couriers in terms of the temporal-spatial distribution of the deliveries.

The delivery capacities of the UAVs and the couriers are affected by multiple factors such as restaurant and destination locations, parcel occurrence time, the speed and locations of UAVs and couriers, etc, as demonstrated in Sec. 4. Specifically, we calculate the flight distance of the UAVs to deliver the parcels within each grid. Fig. 10 shows the flight distance of the UAVs of one station to deliver the parcels. Most parcels can be delivered with the UAV flight distance within 12km. We also calculate the delivery time of a courier to deliver a parcel during different periods. Fig. 11 illustrates the courier delivery time within the peak hours, i.e., between 11 : 00 and 11 : 30, the couriers can deliver the most instant parcels within 35 minutes. To derive the possible flight distance of UAVs and delivery time by couriers, we feed the obtained UAV flight distance and courier delivery time into a Kernel Density Estimation (KDE) model. The Probability Density Functions (PDF) of the distributions of UAV flight distance and the delivery time by couriers can be inferred, respectively, as shown by the fitting curves in Fig. 10 and Fig. 11. According to the red dash lines in Fig. 10, the UAVs can fly for 24.1km and 17.0km carrying the parcel of 1kg and 2kg, respectively. Both distance is much longer than all the obtained UAV delivery distances, as shown in Fig. 10. The KDE model is as Eq.(16).

$$\hat{f}(L) = \frac{1}{mh} \sum_{k=0}^{m-1} K\left(\frac{L - L_k}{h}\right), -\infty < l < +\infty, \quad (16)$$

where m is the total sample number of the historical flight distance or delivery time by couriers, L_k is the flight distance or the delivery time of the k -th sample, h represents the smoothing parameter impacting the estimation accuracy of the model, which can be obtained by the Mean Integrated Squared Error (MISE) criterion. The Gaussian function is leveraged as the kernel function $K(\cdot)$, which is a monotonically decreasing function of the

flight distance or the delivery time L [77]:

$$K\left(\frac{L-L_k}{h}\right) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{(L-L_k)^2}{2h^2}\right]. \quad (17)$$

The in-advance recruited couriers participate in the delivery in the next time period ΔT , i.e. 30 minutes or 1 hour. The predicted delivery demand occurring in the following ΔT at the grid g is denoted as $\tilde{\mathbb{D}}(g)$. Let $\hat{f}_g(L)$ denote the estimated PDF of the UAV flight distance within a grid g and $\hat{f}(t)$ denote the courier delivery time PDF in the following ΔT . After one delivery, the UAV will replenish its energy, such as swapping its battery. Let τ denote the battery replenishment time. $\mathbb{E}$ is the set of UAV charging stations. To sum up, the required courier number UAVs at the time t is determined as:

$$N_c(t) = \sum_{e \in \mathbb{E}} \frac{\left(\sum_{g \in \mathbb{G}(e)} \left| \tilde{\mathbb{D}}(g) \right| \times \left(\frac{\hat{f}_g(L)}{v(u)} + \tau \right) - |\mathbb{U}(e)| \times \Delta T \right) \times \text{mean}_{g \in \mathbb{G}(e)} \frac{v}{\int_0^{+\infty} L \hat{f}_g(L) dL}}{\Delta T} \times \hat{f}(t), \quad (18)$$

where $\left(\frac{\hat{f}_u(g)}{v} + \tau \right)$ is the UAV delivery time for an instant parcel from the grid g ; $\sum_{g \in \mathbb{G}} \left(\left| \tilde{\mathbb{D}}(g) \right| \times \left(\frac{\hat{f}_u(g)}{v} + \tau \right) \right)$ is the needed UAV delivery time for delivering all predicted parcels near the charging station e ; $|\mathbb{U}(e)| \times \Delta T$ is the total delivery time that can be provided by the UAVs in station e during ΔT ; $\text{mean}_{g \in \mathbb{G}(e)} \frac{\int_0^{+\infty} L \hat{f}_g(L) dL}{v}$ denotes the average delivery time of UAV near station e . The number of parcels that can be delivered by a courier during ΔT is $\frac{\Delta T}{\hat{f}(t)}$. $N_c(t) \leq 0$ indicates the UAVs in the station e can deliver all the predicted parcels ordered near e in the following ΔT , and crowdsourced couriers are not necessary. On the contrary, $N_c(t) > 0$ indicates the UAVs can only deliver a part the predicted instant parcels, and $|N_c(t)|$ couriers should be recruited.

5.3 Optimization Problem Formulation

Let the optimal variables $x(u, p)$ and $x(c, p)$ respectively represent the delivery task assignment strategy: $x(u, p) = 1$ indicates the instant parcel $p \in \mathbb{P}$ is delivered by the UAV u , and $x(u, p) = 0$, otherwise; $x(c, p) = 1$ denotes p is delivered by the courier c , and $x(c, p) = 0$, otherwise. Given $x(u, p)$ and $x(c, p)$, $\mathbb{P}_u$ and $\mathbb{P}_c$ can be respectively presented as: $\mathbb{P}(u) = \{p | x(u, p) = 1, \forall u \in \mathbb{U} \& p \in \mathbb{P}\}$ and $\mathbb{P}(c) = \{p | x(c, p) = 1, \forall c \in \mathbb{C} \& p \in \mathbb{P}\}$. Then the optimization problem is formulated as:

$$\max f = \sum_{u \in \mathbb{U}, p \in \mathbb{P}} x(u, p) w(p) + \gamma_1 \sum_{c \in \mathbb{C}, p \in \mathbb{P}} x(c, p) (\gamma_2 - w(p)), \quad (19)$$

$$\text{s.t. } x(u, p), x(c, p) \in \{0, 1\}, \forall p \in \mathbb{P}, \forall u \in \mathbb{U}, \forall c \in \mathbb{C}, \quad (20)$$

$$\sum_{u \in \mathbb{U}} x(u, p) + \sum_{c \in \mathbb{C}} x(c, p) \leq 1, \forall p \in \mathbb{P}, \quad (21)$$

$$\mathbb{P}(u) = \{p | x(u, p) = 1, \forall u \in \mathbb{U}, p \in \mathbb{P}\}, \quad (22)$$

$$\mathbb{P}(c) = \{p | x(c, p) = 1, \forall c \in \mathbb{C}, p \in \mathbb{P}\}, \quad (23)$$

$$T(u, \mathbb{P}(u)) \text{ and } T(c, \mathbb{P}(c)) \text{ are feasible}, \forall u \in \mathbb{U}, \forall c \in \mathbb{C}. \quad (24)$$

The optimization problem is explained as follows. The f function in Eq.(19) is the optimization objective, which is in fact a multi-objective function. The first term $\sum_{u \in \mathbb{U} \& p \in \mathbb{P}} x(u, p) w(p)$ in f is to maximize the total reward (cost) of the parcels assigned to the UAVs, where $w(p)$ is the reward (cost) of delivering a parcel earned by the courier (paid by the company). Since the aBeacon dataset includes no reward information, this work randomly samples $w(p)$ between 1 and 3. The second term refers to the performance of couriers. Specifically, γ_1 is the weight of the second term, similar to the weights for different objectives of a typical multi-objective problem.

Since the proposed UCD is company-oriented, which should reduce its paid reward (cost). So, for a parcel p with its reward $w(p)$ being larger than the threshold γ_2 , $x(c, p)$ should be 0 in any case; while if $w(p)$ is smaller than the threshold, $x(c, p)$ may (but not guaranteed to) be 1. However, to guarantee the income of couriers, the objective function f does not mean the company should not always assign the parcel with the lowest reward (cost) to the couriers. The value of γ_2 balances the profit between couriers and the express company: Given a larger γ_2 , there would be more candidate parcels with $\gamma_2 - w(p) > 0$ that can be assigned to the courier c , namely $x(c, p)$ may be 1, which increases the income of couriers; with a smaller γ_2 , there would be fewer candidate parcels that can be delivered by couriers, which decreases the income of couriers but increases the profit of the company. Clearly, the value of γ_2 cannot be arbitrarily small, otherwise, the delivery capacity of UCD cannot satisfy the demand and the couriers would not participate to the tasks of the company in the future. The binary optimal variables $x(u, p)$ and $x(c, p)$ are stated in Eq.(20). For any instant parcel $p \in \mathbb{P}$, it can be assigned to one UAV or one courier, which is guaranteed by Eq.(21). The definitions of $\mathbb{P}(u)$ and $\mathbb{P}(c)$ in Eq.(22) and Eq.(23) claims the relationships of $\mathbb{P}(u)$ and $\mathbb{P}(c)$ with $x(u, p)$ and $x(c, p)$. Eq.(24) claims that the delivery paths of both the UAVs and the couriers should be feasible. The problem is complex and at least np-hard, because the instant delivery task assignment problem involves the well-known subset selection problem Eq.(20) and (21), and the Traveling Salesman Problem Eq.(24) in generating delivery routes for UAVs and couriers, which are both np-hard. A heuristic algorithm is described in Sec. 5.4.

5.4 Optimization Problem Solution

The delivery task assignment algorithm is presented in Alg. 1. The algorithm first initializes the objective variables in line 1. The algorithm iteratively executes until all the assigned parcels of the UAVs and couriers reaches their delivery capacity or all parcels are assigned (Line 2). In each loop, for any UAV $u \in \mathbb{U}$, which has not yet reach its delivery capacity, schedules the new feasible path $T(u, \mathbb{P}(u) \cup \{p\})$ for every unassigned parcels $p \in \mathbb{P}$ using the heuristic principle of [81](Line 3-6). If a UAV cannot schedule a feasible path with one more parcel, the UAV has reached its delivery capacity and should not be scheduled to deliver more parcels (Line 7-8). Otherwise, obtain the parcel $p(u)$ for each UAV u with the maximum marginal gain: $\arg \max_p \Delta f(u) = \arg \max_p \frac{w(p)}{\Delta t(u, p, \mathbb{P}(u))}$ (Line 9-11). The operations (Line 13-22) for each courier c are similar to the operation for each UAV u (Line 4-12), except that the maximum marginal gain for each courier c is $\arg \max_p \Delta f(c) = \arg \max_p (\gamma_1 \gamma_2 - \gamma_1 \frac{w(p)}{\Delta t(c, p, \mathbb{P}(c))})$. If the maximum marginal gain of the UAVs ($\max \Delta f(u)$) is larger than the maximum marginal gain ($\max \Delta f(c)$) of the couriers, the parcel $p(u)$ is assigned to the UAV u (Line 23-25); Otherwise, the parcel $p(c)$ is assigned to the courier c for delivery (Line 26-28).

6 EVALUATION

In this section, we conduct a data-driven evaluation using the real-world datasets introduced in Sec. 3 to demonstrate the efficiency of the proposed UCD framework.

6.1 Experimental Settings

The one-month datasets of *aBeacon* and the contextual datasets are divided into two parts: the first 23-day data is utilized to train the prediction neural network model of UCD, while the last 7 days data is used to evaluate the performance of our approach. Since the *aBeacon* dataset provides the spatial distribution of couriers but without detailed position information, we randomly generate the positions for each courier under the constraint of known spatial distribution.

Additionally, four important factors that may dramatically impact delivery performance are considered in the evaluation: delivery demand, UAV charging station number, UAV number per charging station, and courier capacity. Table 2 shows the default value of each parameter and the motivation of all the default values. According

Algorithm 1 The Delivery Assignment Algorithm in UCD.

Input: The undelivered parcel set $\mathbb{P}$, the UAV set $\mathbb{U}$, the courier set $\mathbb{C}$, the locations of $u \in \mathbb{U}$.

Output: The delivery strategy $x(u, p)$ and $x(c, p)$, the feasible delivery paths of couriers and UAVs $T(c, P(c))$ and $T(u, P(u))$.

```

1: Initialize  $x(c, p) = x(u, p) = 0, \forall c \in \mathbb{C}, \forall u \in \mathbb{U}, \forall p \in \mathbb{P}$ ;
2: while  $\mathbb{U} \neq \emptyset \& \mathbb{P} \neq \emptyset$  or  $\mathbb{C} \neq \emptyset \& \mathbb{P} \neq \emptyset$  do
3:   for  $u \in \mathbb{U}$  do
4:     for  $p \in \mathbb{P}$  do
5:       Generate the feasible delivery path  $T(u, \mathbb{P}(u) \cup \{p\})$  using the heuristic principle [81];
6:     end for
7:     if  $T(u, \mathbb{P}(u) \cup \{p\})$  is infeasible,  $\forall p \in \mathbb{P}$  then
8:       Delete  $u$  from  $\mathbb{U}$ ;
9:     else
10:      Obtain  $p(u) = \arg \max_p \Delta f(u) = \arg \max_p \frac{w(p)}{\Delta t(u, p, \mathbb{P}(u))}$ ;
11:    end if
12:  end for
13:  for  $c \in \mathbb{C}$  do
14:    for  $p \in \mathbb{P}$  do
15:      Generate the feasible delivery path  $T(c, \mathbb{P}(c) \cup \{p\})$  using the heuristic principle [81];
16:    end for
17:    if  $T(c, \mathbb{P}(c) \cup \{p\})$  is infeasible,  $\forall p \in \mathbb{P}$  then
18:      Delete  $c$  from  $\mathbb{C}$ ;
19:    else
20:      Obtain  $p(c) = \arg \max_p \Delta f(c) = \arg \max_p (\gamma_1 \gamma_2 - \gamma_1 \frac{w(p)}{\Delta t(c, p, \mathbb{P}(c))})$ ;
21:    end if
22:  end for
23:  if  $\max \Delta f(u) > \max \Delta f(c)$  then
24:    Set  $x(u, p(u)) = 1 \& \mathbb{P}(u) = \mathbb{P}(u) \cup \{p(u)\}$ , where  $u = \arg \max_u \Delta f(u)$ ;
25:    Delete  $p(u)$  from  $\mathbb{P}$ ;
26:  else
27:    Set  $x(c, p(c)) = 1 \& \mathbb{P}(c) = \mathbb{P}(c) \cup \{p(c)\}$ , where  $c = \arg \max_c \Delta f(c)$ ;
28:    Delete  $p(c)$  from  $\mathbb{P}$ ;
29:  end if
30: end while

```

to the table, there are 200 UAVs operating in the city area about $6,400 \text{ km}^2$. In the following, the underlined value of each factor is the default value in the evaluation.

- **UAV Charging Station Number:** The value can be sampled from [10, 15, 20, 25]. Station locations are determined through K-means++ over the distributions of historical instant deliveries.
- **UAV Number per Charging Station:** The value can be sampled from [1, 5, 10, 15], which indicates the consistent number of UAVs in each station.
- **Courier Capacity:** It denotes the maximum number of parcels a courier can deliver and is sampled from [1, 2, 3, 4, 5]. For simplicity, only the simultaneously carried parcel number is considered in the evaluation, since we cannot capture the weight limit of a courier.

Table 2. Default Values and Motivation.

Parameter	Default Value	Motivation
# of UAV Charging Stations	20	With this number of Stations, UAVs can deliver more than half of the parcels.
# of UAVs in Each Station	10	With this number of UAVs, the UAVs can deliver more than half parcels.
Courier Capacity	3	With this value, each courier can deliver almost his assigned parcels within the delivery time limit.
Delivery Demand	1	The value indicates that all parcels to be delivered to evaluate the performance.
γ_1, γ_2	7.5, 1.75	Obtained by Evaluation (See Fig. 12 and 13).

- **Delivery Demand:** The values can be sampled from $[0.7, 0.8, 0.9, 1]$, where each value represents a proportion of the instant delivery orders among the original orders aBeacon are delivered cooperatively by UAVs and couriers.

6.2 Metrics

To evaluate the delivery performance in terms of UAVs and by couriers, we exploit the following three metrics.

- **Delivered Parcel Number** is the number of instant parcels delivered by the cooperative delivery method, including those delivered by both UAVs and couriers. This metric is one of the basic metrics for delivery.
- **Delivery Time** indicates the average time from the instant parcels being ordered to the parcels being sent to the customers. This metric is another basic metric of parcel delivery.
- **Parcel Delivered by Each Courier** indicates the number of the instant parcels dispatched to the courier when he is recruited. Since the couriers are paid for delivery, this metric is vital for couriers.

6.3 Baselines

Since UCD represents the initial exploration of cooperative instant delivery involving UAVs and couriers, no pre-existing baseline methods are available. To assess UCD's effectiveness, we compare UCD against two heuristic baselines and the practical solution extracted from the aBeacon data-set.

- **Only UAV (O-U).** The UAVs may be insufficient and can only deliver part of the parcels. Following the well-known (single-)subset selection problem with route constraint [3, 81], the problem can be modeled as a multi-subset selection problem with route constraint. O-U leverages the same heuristic principle with the heuristic algorithm [81] to solve the problem of O-U.
- **UCD-G.** Intuitively, the UAVs should deliver the long-haul parcels and leave the short-haul ones to the couriers. The UCD-G baseline always schedule UAVs to deliver the longest parcels. The number of recruited couriers is determined using 1) historical data for delivery demand estimation, and 2) the courier recruitment method in Sec. 5.2.
- **Practical Solution from aBeacon (P-S)**[16]. This baseline is a commercial delivery method, which hires human couriers only for delivery. The performance of P-S is extracted from the aBeacon dataset [16].

6.4 Numerical Results

6.4.1 The impacts and values of γ_1 and γ_2 . Fig. 12 and Fig. 13 respectively illustrate the impacts of values of γ_1 and γ_2 on the parcel assignment. The x-axes indicate time periods during the day, with each slot representing

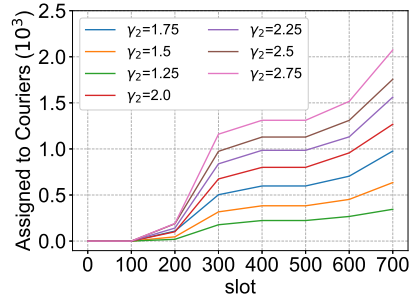
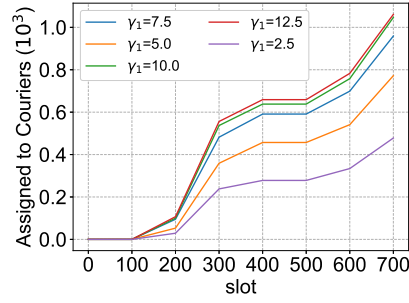


Fig. 12. Impacts of γ_1 on Parcel Assignment. Fig. 13. Impacts of γ_2 on Parcel Assignment.

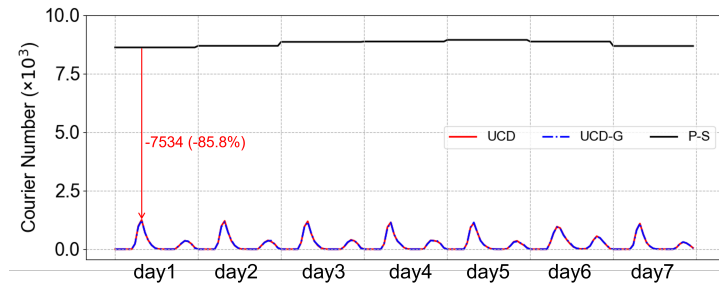


Fig. 14. Courier Recruitment Number by UCD, UCD-G and P-S.

one minute. The parcel assignment here only occurs for those parcels that can be delivered by both UAVs and couriers. As in Fig. 12, the value of γ_1 shows a significant impact on delivery assignment. When γ_1 is 2.5, the parcels assigned to the couriers reach the lowest: 478. By contrast, the couriers can deliver over 1,000 parcels when γ_1 is larger than 10. Besides, there are two peaks in this figure, which happens around the slot 400 and slot 700. They are respectively after two dining periods during a day, when the UAVs have a heavy delivery burden. To reach a balance between delivered parcel number by couriers and delivery efficiency of UAVs. We finally set $\gamma_1 = 7.5$, which ensures a relatively high delivery efficiency of the UAVs and the couriers. Moreover, Fig. 13 visualize how γ_2 affects assigned parcel number. The larger value of γ_2 is, the more parcels assigned to couriers will be. If the couriers are allowed to ship parcels which are as precious as 2.75 and can be shipped by UAVs, 2,074 instant parcels will be delivered by couriers in this case. In contrast, only 977 parcels are assigned to couriers when γ_2 is only 1.25. Therefore, We set the default value of γ_2 as 1.75, considering couriers' reward and delivery efficiency.

6.4.2 Courier Recruitment Number and Assigned Parcel Number to UAVs. The recruited courier numbers of different approaches under the default parameters are illustrated in Fig. 14. The number of dedicated couriers recruited by P-S remains stable each day (as shown by the black curves), while UCD and UCD-G can recruit couriers on-demand. In Fig. 14, the red dash line insights gap of the maximum recruited courier number between UCD and P-S, indicating UCD can reduce the employment cost for delivery companies. Specifically, UCD reduces the courier number by at least 7,434 (85.8%) compared to P-S; namely, the maximum needed courier number of UCD is only 14.2% of P-S. Additionally, there are two peaks (noon peak and afternoon peak) of courier number in UCD and UCD-G, which correspond to the two parcel peaks as shown in Fig. 3(c). UCD and UCD-G recruit

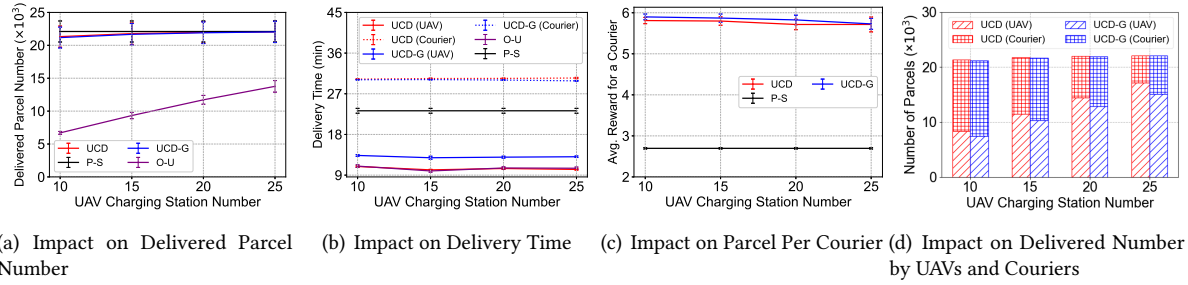


Fig. 15. Impact of UAV Charging Station Number on Delivery Performance.

about 1,250 couriers to deliver the instant parcels during the noon peak, while only 300 of couriers are needed to deliver the instant parcels during the ordinary time in UCD and UCD-G, respectively. By contrast, the P-S recruits over 8,600 dedicated couriers for delivery during the 7 days. This is because the participation of UAVs in parcel delivery reduces the demand for couriers. The extremely small number of required couriers in UCD not only reduces delivery costs but also alleviates the impacts of insufficient delivery capacity caused by unstable and inadequate numbers of available couriers on the crowdsourcing platform. Besides, this figure shows that the required couriers is time-varying. If the number of recruited couriers is still insufficient during peak hours, several potential solutions can be employed. First, the incentive reward can be increased to encourage more couriers to perform better [70]. Increasing the cost is more feasible in cooperative UAV-courier delivery because the number of required couriers is very small, and the total increase in cost would be limited. Second, Algorithm 1 can assign the UAVs to deliver the parcels with shorter delivery time ($t(u, p)$), by reducing the weight parameter γ_2 , thus each UAV can deliver more parcels and the company requires fewer couriers.

6.4.3 Impact of UAV Charging Station Number. The impacts of the UAV charging station number on the delivery performance are illustrated in Fig. 15.

The number of UAVs increases with the growth in the number of charging stations. Thus, the delivered parcel numbers of all methods except for P-S grow as shown in Fig. 15(a). UCD delivers almost all (99.8%) parcels with 25 UAV charging stations in the city, which is similar with that of UCD-G and higher than O-U by 37.6%, respectively. This is because crowdsourced couriers are involved in both UCD and UCD-G. Due to their high speed and disregard for ground traffic, UAVs have much shorter delivery times compared to couriers. The more UAVs available for delivery, the less pressure on each individual UAV, which reduces the UAV delivery time, as illustrated in Fig. 15(b). Moreover, the UAV delivery time in UCD is shorter than that in UCD-G by 21.5%, because UAVs in UCD-G are tasked with delivering parcels over longer distances from restaurants to customers. Due to the same courier delivery model in UCD and UCD-G, the delivery time of couriers are similar in UCD and UCD-G, which is $2.97\times$ longer than that of UAVs.

Although the increase in the number of UAVs leads to fewer parcels available for couriers to deliver, the couriers in UCD and UCD-G still earn much more reward than in P-S by factors of $2.12\times$ and $2.19\times$, respectively, according to Fig. 15(c). This is because the UAVs help to deliver the parcels too far away or with too low reward. Besides, as the salary restriction in UCD, more UAVs are assigned to deliver parcels compared to that in UCD-G, and further increases the number of parcels which can only be shipped by couriers. This leads to similar rewards for couriers in UCD and UCD-G as shown in Fig. 15(c). Fig. 15(d) illustrates the proportion of delivered parcels by UAVs and by couriers. 8,328 (39.0%) and 7,436 (34.8%) of instant parcels are delivered by UAVs when there are only 1 UAV charging stations in UCD and UCD-G, respectively. When the number of UAV charging stations reaches 25, the UAVs respectively deliver 17,180 (77.9%) and 15,128 (68.6%) instant parcels in UCD and UCD-G.

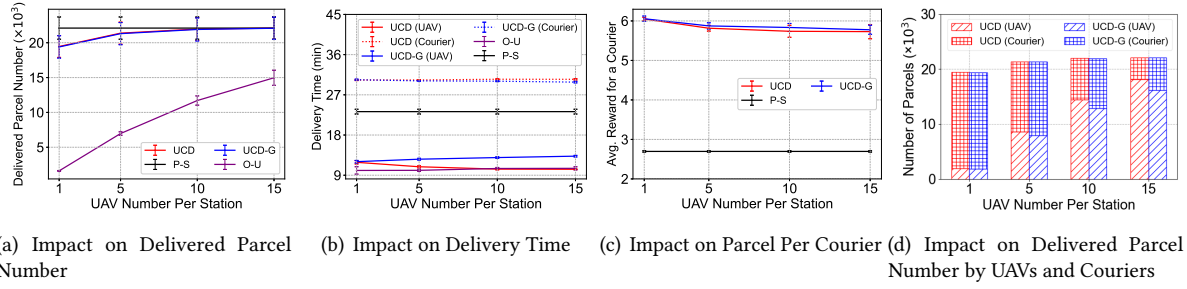


Fig. 16. Impact of UAV Number Per Station on Delivery Performance.

This is because, to improve delivery efficiency for couriers, UAVs in UCD-G are assigned to deliver parcels over longer distances, which leads to fewer parcels delivered by the UAVs.

6.4.4 Impact of UAV Number Per Station. Fig. 16 shows the impact of UAV number per charging station on the delivery performance.

As illustrated in Fig. 16(a), the delivered instant parcel number in UCD grows by 13.5% when the UAV number at each station increases from 1 to 15, and the number of parcels delivered by O-U soars from 1,614 to 14,971. The two methods with couriers participating catch up with P-S, in which nearly all parcels are delivered when there are 10 UAVs at each station. This is because the couriers greatly improve the overall delivery capacity in these methods. Besides, the delivery efficiency is also improved with more UAVs available for parcel delivery, which leads to the downward trend of UAV delivery time from 11.9 minutes to 10.3 minutes in UCD as shown in Fig. 15(b). More importantly, the UAV delivery efficiency of UCD surpasses that of O-U when there are 10 UAVs in each station. This is because the parcels assigned to the UAVs in UCD are all effective for delivery. The reason why the delivery time by the couriers in P-S is much shorter (−31.2%) than those in UCD and UCD-G is that the parcels with short delivery distance and high efficiency are shipped by the UAVs.

Fig. 16(c) plots the average reward obtained by each courier. Since more UAVs can deliver the parcels that would be delivered by couriers if there were no sufficient UAVs, the parcels shipped by the couriers are more ineffective. This results in slight decreases of the courier reward by 5.0% in UCD and 4.6% in UCD-G, respectively. However, the couriers in UCD gain much more reward (113.0%) than that in P-S. Fig. 16(d) illustrates the impact of the UAV number per station on the number of parcels delivered by the UAVs and the couriers, respectively. Only 1,948 (10.0%) and 1,901 (9.4%) of the parcels are delivered by the UAVs respectively in UCD and UCD-G, when there is only one UAV in each station. When the UAVs in each charging station are 15, about 18,100 (81.9%) of parcels are delivered by the UAVs in UCD, and that in UCD-G reaches 16,124 (72.9%). This is because the delivery reward (cost) by the UAVs is lower than the delivery reward (cost) by couriers.

6.4.5 Impact of Courier Capacity. Fig. 17 illustrates the impact of courier capacity on the delivery performance. The increase in courier capacity means more parcels can be delivered by a courier simultaneously, which decreases the recruited courier number. The curves of O-U are stable due to the absence of couriers.

As illustrated in Fig. 17(a), the number of the delivered parcels in UCD grows by only 2.6% when the courier capacity increases from 1 to 5. This is because more couriers are hired when they cannot carry more parcels simultaneously. Besides, when the courier capacity reaches 5, 22,013 of the parcels are delivered in UCD, which is 88.2% more than that in O-U. Fig. 17(b) plots the influences of the courier capacity on the delivery time. The Delivery time by the UAVs remains stable with the increase of courier capacity. This is because the parcel assignment to UAVs are not affected by courier delivery capacity. By contrast, the delivery time of the couriers skyrockets with the increasing courier capacity (+50.5%), especially when it grows from 1 to 5 (+39.4%). This is

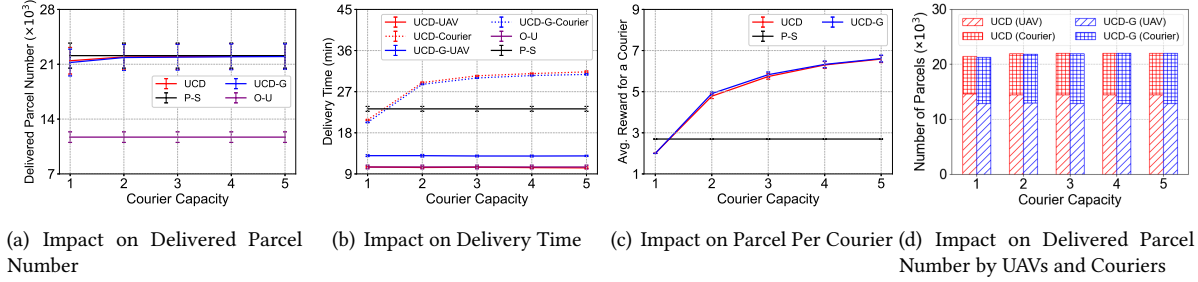


Fig. 17. Impact of Courier Capacity on Delivery Performance.

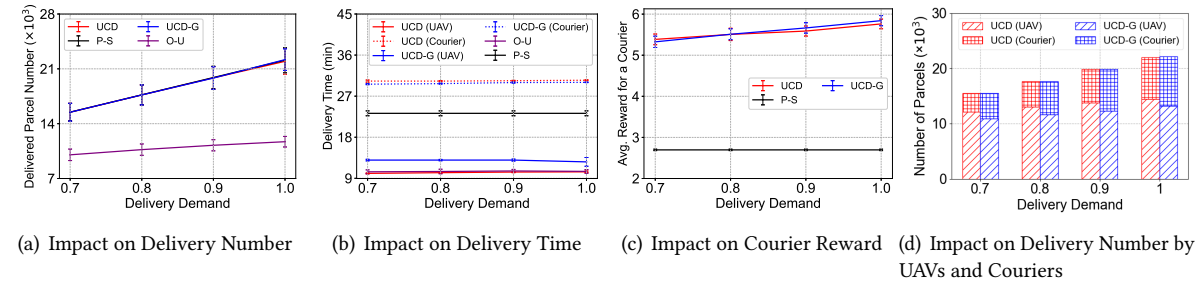


Fig. 18. Impact of Delivery Demand on Delivery Performance.

stemmed from the couriers tend to pick up the new parcels rather than drop off the carrying parcels to obtain more reward when their delivery capacity increases. Thus, the drop-offs of the carrying parcels are postponed and the delivery time increases.

The increase in the courier capacity directly improves the reward of the couriers as shown in Fig. 17(c). The couriers in UCD only earn 2.0 when they are allowed to carry one parcel at the same time, while their reward soars to 6.58 if they can carry 5 parcels simultaneously. Compared to the couriers' reward in P-S, it is a high-paying job (+144.6% of reward ascending) for the couriers in UCD. Fig. 17(d) shows the delivered parcel number by UAVs and couriers, respectively. The UAVs deliver about 14,562 and 12,911 of instant parcels respectively in UCD and UCD-G with the increase in courier capacity. This is because the increase in courier capacity does not affect the delivery by UAVs. Besides, due to the considerable UAV delivery capacity, the number of parcels delivered by couriers does not rise too much, only accounting for 11.8% in UCD and 9.3% in UCD-G.

6.4.6 Impact of Delivery Demand. Fig. 18 illustrates the impacts of the delivery demand on the delivery performance.

Fig. 18(a) shows the impacts of the delivery demand on the delivered parcel number by UCD and baselines. The delivered parcel number of P-S is shrunk according to the discounts of the delivery demand. The parcels delivered in UCD increase by 42.0% when the demand increases from 70% to 100%. More than 21,950 parcels are delivered by UCD with all parcels available, which is respectively 0.2% and 87.7% higher than UCD-G and O-U. Besides, UCD and UCD-G can always deliver almost all of the parcels. The effects of the delivery demand on the delivery time are illustrated in Fig. 18(b). As the UAVs and couriers are sufficient to deliver all available parcels, the increase of the delivery demand affects the delivery efficiencies of UAVs and couriers in all methods slightly. The UAVs in UCD achieve shorter delivery time (−19.8%) compared to that in UCD-G, while the courier delivery time in these two methods is relatively close (varying only 1.5%). Because the UAVs deliver the parcels that are

effective to ship, the delivery times by couriers in UCD and UCD-G are 31.2% and 29.2% longer than that in P-S, respectively.

Fig. 18(c) plots the impact of the delivery demand on couriers' reward. A higher reward (7.1%) is obtained by each courier with the increase of delivery demand in UCD. Besides, once the delivery demand is 100%, the couriers gain more reward (+114.1%) in UCD compared to that in P-S. This is caused by that the parcels with longer detour distance for the couriers are delivered by the UAVs considering delivery efficiency. Fig. 18(d) shows the number of the parcels delivered by the UAVs and the couriers in UCD and UCD-G. More parcels are available to be delivered, more parcels are delivered by the couriers, and the number of parcels delivered by UAVs in the two methods stays stable. This is because the delivery capacity of the UAVs is fulfilled when the delivery demand is 0.7. Since UCD-G assigns parcels with longer delivery distance to UAVs, more parcels (+12.1% with all parcels available) are delivered by UAVs in UCD than in UCD-G.

7 CONCLUSION

This paper investigates the first air-ground cooperative delivery framework named UCD, using UAVs and crowdsourced couriers. The design of UCD is based on multi-source data, which aims to fill the delivery capacity gap of delivery UAVs by crowdsourcing an appropriate number of part-time couriers. The delivery models of the UAV and of courier are formulated. Besides, a data-driven method is employed for delivery demand prediction. Then the recruiting courier number is determined using both the delivery models of the UAV and courier, and the predicted instant delivery demand. After that, the instant delivery tasks are assigned between UAVs and the couriers based on their delivery status. Finally, the proposed UCD is comprehensively evaluated using a set of long-term, real-world data-sets. Evaluation results demonstrate the superior performance of the proposed method. However, despite the superior performance, there are still some research directions for UCD that need more investigation. Firstly, there are still some unavailable data, such as the precise distribution data or model of the working hours of the part-time couriers, which have not been fully taken into account and may result in a more precise courier number determination and efficient delivery task assignment. In addition, although there have been multiple works on the locations of UAV stations, such as [35, 54, 69], these works mainly consider the delivery model of UAVs and the spatial features of delivery demand. With the spatial-temporal data of the part-time couriers, the impact of the couriers to the UAV stations' location can be utilized to optimize the locations of the UAV stations in UCD. Moreover, the current design does not systematically take into account of the unmanned ground vehicle (UGV) for the non-flying zones of the city. Cooperative courier-UAV-UGV delivery considering the non-flying zones can extend the scalability and adaptability of the cooperative delivery paradigm.

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Appendix A FLIGHT PATH PLANNING CONSIDERING GROUND BARRIERS

This appended section presents the flight path planning algorithm for a UAV to avoid collisions with obstacles and terrains, such as skyscrapers in the city. Let h indicate the cruising altitude of UAVs. After picking up instant parcels, UAVs rise to h and remains at that altitude until they reach destination cabinets. Thus, the buildings higher than h are the obstacles to be avoided by UAVs. This path planning algorithm is inspired by the sampling-based algorithm proposed by Lin et al. [43].

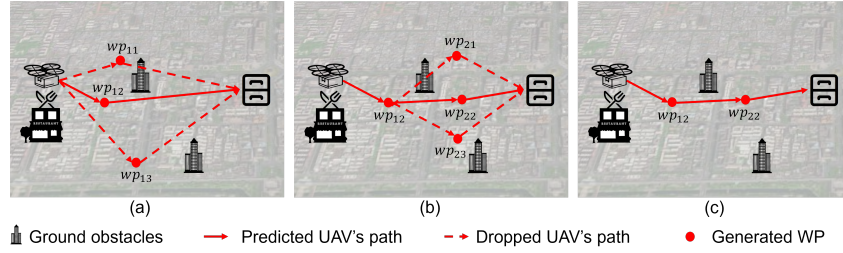


Fig. 19. UAV Flight Path Generation by Sampling-based Algorithm. (a) Basic Idea of The Sampling-based Algorithm; (b) Iterative Sampling; (c) The Scheduled Collision-Free Path.

Specifically, when a UAV plans its flight path, it senses the terrains and obstacles on its way using the digital map and its on-board sensors. Then, the sampling-based algorithm will schedule the shortest collision-free flight path by iteratively sampling multiple waypoints (WPs). Fig. 19 illustrates the scheduling process of a flight path from the restaurant to the destination cabinet. When the UAV picks up the parcel and heads to the destination cabinet, the two skyscrapers in Fig. 19 are first sensed. Several WPs (e.g., wp_{11}) are randomly sampled in the free airspace. The UAV can reach the destination cabinet via wp_{12} without colliding with the skyscrapers, as shown in Fig. 19(a). In the figure, the red solid lines represent the predicted flight path, and the red dotted lines indicate the paths discarded due to long flight time or collision. Because there are additional obstacles between wp_{12} and the destination cabinet, other WPs (e.g., wp_{21}) are sampled, as shown in Fig. 19(b). Once wp_{22} is chosen, there are no remaining terrains or obstacles to the destination cabinet, allowing the UAV to fly directly to the destination without sampling any additional WPs, as illustrated in Fig. 19(c).